In the case of Langevin dynamics, we identify the vectors $\boldsymbol{b}_i$ in (6.43) with the terms in (6.47). One puts Langevin dynamics into the correct form by using

$$\boldsymbol{b}_0 = \begin{bmatrix} \boldsymbol{p} \\ -\nabla U(\boldsymbol{q}) - \gamma \boldsymbol{p} \end{bmatrix}, \quad \boldsymbol{b}_i = \sigma \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{e}_i \end{bmatrix}, \quad i = 1, 2, \ldots, \mathrm{N_c},$$

where $\boldsymbol{e}_i$ is the ith Euclidean basis vector in $\mathbb{R}^{\mathrm{N_c}}$ and $\sigma = \sqrt{2\gamma \mathrm{k_B T}}$ is constant. Now it is a straightforward matter of computing the commutators. For $i = 1, 2, \ldots, \mathrm{N_c}$, we have, using the fact that $\boldsymbol{b}_i$ is a constant vector field and so its Jacobian matrix vanishes,

$$[\boldsymbol{b}_0, \boldsymbol{b}_i] = -\begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -U''(\boldsymbol{q}) & -\gamma \boldsymbol{I} \end{bmatrix} \boldsymbol{b}_i = -\sigma \begin{bmatrix} \boldsymbol{e}_i \\ -\gamma \boldsymbol{e}_i \end{bmatrix},$$

where $U''(\boldsymbol{q})$ is the Hessian matrix of U. As these $\mathrm{N_c}$ vectors, taken together with the original $\boldsymbol{b}_i$, clearly provide a linearly independent set of $2\mathrm{N_c}$ vectors, we have verified Hörmander's condition. Thus Assumption 1(ii) holds in the case of Langevin dynamics.

It remains to show that Assumption 1(i) is satisfied. We will in fact show a stronger result, given in the following lemma:

Lemma 6.1 *For all $t > 0$ and all open $C \subset \mathcal{D}$, we have that*

$$\Pr\left(\mathbf{X}(t) \in C \,|\, \mathbf{X}(0) = \mathbf{X}_0\right) > 0, \quad \forall \mathbf{X}_0 \in \mathcal{D},$$

where $\mathbf{X}(t) = (\boldsymbol{q}(t), \boldsymbol{p}(t))^T$ is a solution to Langevin dynamics (6.47).

Proof It is sufficient to show that the open ball of radius δ centered on a point $\boldsymbol{y} \in \mathcal{D}$, denoted $\mathcal{B}_\delta(\boldsymbol{y}) \subset \mathcal{D}$, has positive probability at any $t > 0$. To show this, the goal is to demonstrate that we have a positive probability of the realization of the noise process taking our initial condition inside the ball.

Consider the ODE

$$\frac{\mathrm{d}^2 \boldsymbol{q}}{\mathrm{d}t^2} + \nabla U(\boldsymbol{q}) + \gamma \frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}t} = \sigma \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}t}, \tag{6.49}$$

for some positive constants γ and σ. This is precisely Langevin dynamics (6.47), substituting a process $\boldsymbol{R}(t)$ in place of the Wiener process. Choose $\boldsymbol{q}$ to be a C^∞ path such that for a given $t > 0$

$$(\boldsymbol{q}(0), \dot{\boldsymbol{q}}(0))^T = \mathbf{X}_0, \quad (\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))^T = \boldsymbol{y}.$$

Taking $\boldsymbol{R}(0) = 0$, we can obtain a C^∞ function $\mathrm{d}\boldsymbol{R}/\mathrm{d}t$ by substitution.